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Silicide-phase evolution and platinum redistribution during silicidation of $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ specimens

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Silicide-phase evolution and platinum redistribution during silicidation of $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ specimens

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We investigated the temporal evolution of nickel-silicide phase-formation and the simultaneous redistribution of platinum during silicidation of a 10 nm thick $\text{Ni}_{0.95}\text{Pt}_{0.05}$ film on a $\text{Si}(100)$ substrate. Grazing incidence x-ray diffraction (GIXRD) and atom-probe tomography (APT) measurements were performed on as-deposited films and after rapid thermal annealing (RTA) at 320 or 420 °C for different times. Observation of the Ni_2Si phase in as-deposited films, both with and without platinum alloying, is attributed to surface preparation. RTA at 320 °C for 5 s results in the formation of the low-resistivity NiSi intermetallic phase and nickel-rich phases, Ni_2Si and Ni_3Si_2 , as demonstrated by GIXRD measurements. At 420 °C for 5 s, the NiSi phase grows outward from the silicide/ $\text{Si}(100)$ interface by consuming the nickel-rich silicide phases. On increasing the annealing time at 420 °C to 30 min, this reaction is driven towards completion. The nickel-silicide/silicon interface is reconstructed in three-dimensions employing APT and its chemical root-mean-square roughness, based on a silicon isoconcentration surface, decreases to 0.6 nm with the formation of the NiSi phase during silicidation. Pt redistribution is affected by the simultaneous reaction between Ni and Si during silicidation, and it influences the resulting microstructure and thermal stability of the NiSi phase. Short-circuit diffusion of Pt via grain boundaries in NiSi is observed, which affects the resultant grain size, morphology, and possibly the preferred orientation of the NiSi grains. Pt segregates at the $\text{NiSi}/\text{Si}(100)$ heterophase interface and may be responsible for the morphological stabilization of NiSi against agglomeration to temperatures greater than 650 °C. The Gibbsian interfacial excess of Pt at the $\text{NiSi}/\text{Si}(100)$ interface after RTA at 420 °C for 5 s is 1.2 ± 0.01 atoms nm^{-2} and then increases to 2.1 ± 0.02 atoms nm^{-2} after 30 min at 420 °C, corresponding to a decrease in the interfacial free energy of 7.1 mJ m^{-2} . © 2012 American Institute of Physics. [<http://dx.doi.org/10.1063/1.4751023>]

I. INTRODUCTION

NiSi is currently the material of choice for contacting advanced complementary metal-oxide semiconductor (CMOS) transistor devices.¹ Ever decreasing device dimensions place stringent constraints on the dimensions and the maximum resistivity of these silicide contacts.² Of the six silicide phases that are stable at room temperature (Ni_3Si , $\text{Ni}_{31}\text{Si}_{12}$, Ni_2Si , Ni_3Si_2 , NiSi , and NiSi_2), the lowest-resistivity monosilicide (NiSi) phase, 10.5–18 $\mu\Omega$ cm, is the one of greatest interest.³ NiSi source and drain contacts are typically formed employing a self-aligned silicide (salicide) process.⁴ A complex nickel-silicide phase-formation sequence, dependent on many process parameters such as Ni-film thickness,^{5–7} dopant-type and concentration,³ substrate type, annealing conditions, and surface preparation,^{8,9} makes it imperative to understand the solid-state reactions controlling the final phase. Furthermore, the integration of NiSi into current and future CMOS technology is a major challenge due to its low thermal stability. At elevated temperatures (~ 650 – 700 °C), the nickel monosilicide film degrades via either agglomeration or a phase transformation

to the thermodynamically stable, but higher resistivity, NiSi_2 phase.¹⁰ Alloying with nominal concentrations of transition-metal elements, such as Pt, Pd, or Rh, has been shown to improve morphological and phase stability at higher temperatures.^{11,12} We have demonstrated that the transition metals Pd or Pt segregate at the nickel-silicide/ Si interface and they appear to be responsible for enhanced morphological stability.^{13–15} The effects of the reaction between Ni and Si on the simultaneous diffusion of Pt during the silicidation process, and its eventual redistribution in the thin film are not well understood, however. Recent reports on Pt redistribution^{16–23} in nickel silicides concern films much thicker than those relevant for CMOS manufacturing and hence are not discussed herein.

Our experiments were undertaken to understand the kinetics of this reaction-diffusion process²⁴ at different stages of the silicidation process used to form low-resistance contacts. Local-electrode atom-probe (LEAP) tomography,^{25–28} which enables three-dimensional (3D) compositional mapping with sub-nanoscale resolution in direct space, was employed to map the elemental distributions of Ni, Si, and Pt. Additionally, grazing-incidence x-ray diffraction²⁹ (GIXRD) experiments were performed on thin films to determine the crystal structures of the phases formed. We studied 10 nm thick Ni and $\text{Ni}_{0.95}\text{Pt}_{0.05}$ thin-films on $\text{Si}(100)$, using

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GIXRD, in the as-deposited condition and after rapid thermal-annealing (RTA) at 320 °C for 5 s. A second set of thin-films were studied after RTA at 420 °C for 5 s. In a typical manufacturing process, source-drain contact formation is followed by metallization and dielectric insulation processes, which are denoted as the back-end-of-line (BEOL) process.³⁰ To simulate the BEOL thermal budget on silicide phase formation, we studied a set of thin-films after RTA at 420 °C for 30 min; Ni_{0.95}Pt_{0.05} thin-films subjected to RTA treatments at 320 °C (5 s), or 420 °C (5 s or 30 min) were also studied using LEAP tomography.

In this article, we show how NiSi nucleates at the silicide/Si(100) heterophase interface and grows toward the Ni_{1-x}Pt_x/silicide heterophase interface by consuming nickel-rich silicides—Ni₂Si and Ni₃Si₂. Simultaneously, Pt diffuses via short-circuit diffusion along grain boundaries (GBs) in the NiSi phase to reach the NiSi/Si(100) heterophase interface at the early stages of the growth of NiSi. Short-circuit diffusion of Pt has significant implications on the resulting microstructure and leads to smaller NiSi grains with a columnar morphology. The Gibbsian interfacial excess of Pt at the NiSi/Si(100) heterophase interface after RTA at 420 °C for 5 s is 1.2 ± 0.01 atoms nm⁻² and after 30 min at 420 °C it increases to 2.1 ± 0.02 atoms nm⁻². The nickel-silicide/Si(100) heterophase interface is reconstructed in 3D utilizing atom-probe tomography (APT), and its chemical root-mean-square (RMS) roughness decreases with the formation of the NiSi phase to 0.6 nm.

II. EXPERIMENTAL

10 nm thick Ni (control) and Ni_{0.95}Pt_{0.05} (alloy) thin-films were deposited by RF magnetron sputtering onto undoped Si(100) substrates, after removing the native silicon oxide with a 1% dilute hydrofluoric acid solution. Subsequently, 5 nm thick TiN films were deposited as capping layers to prevent oxidation of the samples. Samples were studied in the as-deposited state, and after being subjected to RTA using a lamp-based rapid-thermal processor³¹ at: (i) 320 °C for 5 s; (ii) 420 °C for 5 s; and (iii) 420 °C for 30 min (to simulate the thermal budget of the BEOL process). 3D sub-nanoscale compositional mapping was performed using a Cameca (formerly Imago Scientific Instruments, Madison, WI) LEAP 3000Si tomograph. A pulsed picosecond laser (λ = 532 nm) operating at a pulse repetition rate of 500 kHz with an energy per pulse of 0.5 nJ was used to dissect specimens on an atom-by-atom and an atomic plane-by-plane basis. An evaporation rate of 0.2% ion per pulse was utilized, while the sample was maintained at 40 ± 0.3 K under ultra-high vacuum conditions ($<6.6 \times 10^{-9}$ Pa). APT data were analyzed using Cameca's IVAS program.

Samples for APT were prepared employing the “lift-out”^{32,33} technique using a FEI Helios dual-beam focused-ion-beam (FIB) microscope. A Pt cap layer was deposited onto the TiN capping layer, in the region of interest to minimize Ga⁺ ion implantation and radiation damage, utilizing the in-situ gas-injection system (GIS). This was accomplished employing the electron beam and subsequently the ion beam to deposit 100 and 150 nm of Pt, respectively.

Subsequently, a long wedge-shaped cantilever beam, with a triangular cross-section, was lifted out from the blanket thin-film sample using an Omniprobe³⁴ micromanipulator. This wedge was then cut by the Ga⁺ ion-beam and the resulting smaller pieces of material were attached to a number of Si microposts, residing on a coupon, using the Pt GIS. These cold-welded wedges were next sharpened to a microtip using the Ga⁺ ion-beam and annular mill-patterns to achieve a shank angle of $\sim 15^\circ$ and a local radius of curvature of ~ 50 nm. A 30 kV Ga⁺ ion-beam was used for coarse sharpening and a 5 kV Ga⁺ ion beam was utilized for final sharpening to minimize the radiation damage caused by Ga⁺ implantation. GIXRD measurements were performed at the 5BMC (Dow-Northwestern-DuPont-CAT) beam-line of the Advanced Photon Source at Argonne National Laboratory. A pair of Si(111) monochromators were used to select an energy E of 13 keV (λ = 0.9537 Å) with $\Delta E/E \sim 0.015\%$, where ΔE is the energy spread. The x-ray beam was incident at about twice the critical angle²⁹ of the specimen (0.22°) to enhance the signal from the thin-silicide portion. GIXRD measurements were recorded using a point-detector³⁵ and a MAR 165³⁶ area-detector. Silicon capillary samples were used to calibrate the 2D area measurements, where the sample-to-detector distance was 159.3 mm. Figure 1 displays a 2D area measurement from a blanket Ni_{0.95}Pt_{0.05}/Si(100) thin-film sample subjected to RTA at 420 °C for 5 s. Since we are using a planar sample in grazing incidence geometry, we only capture the azimuthal intensity ($0^\circ \leq \varphi \leq 180^\circ$), with respect to the surface normal, where the diffracted beam is above the sample's surface. Also, visible is the beam

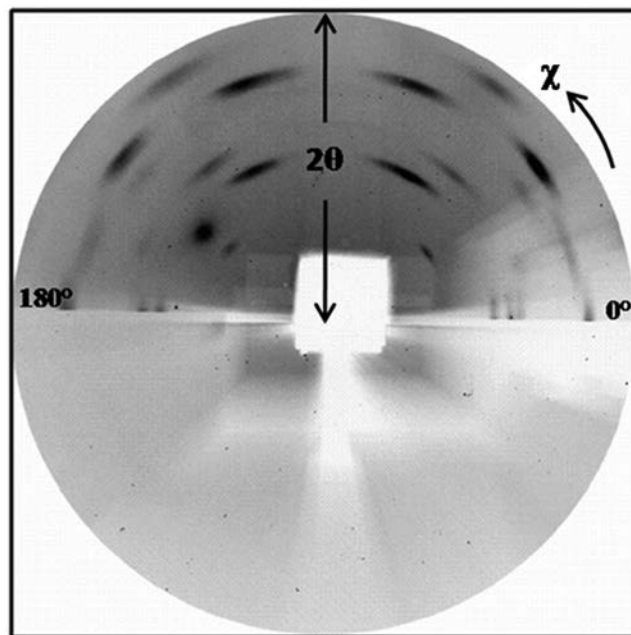


FIG. 1. A gray-scale image of a 2D grazing incidence XRD measurement using an area detector. This image was collected from a Ni_{0.95}Pt_{0.05}/Si(100) sample after annealing at 420 °C for 5 s. Notice the beam stop (white square) at the center of the image. Diffracted intensities above the plane of the thin-film sample are collected over a large azimuthal angle ($0^\circ \leq \varphi \leq 180^\circ$), with respect to the surface normal. Data were analyzed by radially integrating the intensity in small 5° increments of the rotational angle χ on the detector, and plotting it versus 2θ .

stop (white square) at the center of this grey-scale image. Integrating radially over all χ , the rotational angle on the detector, proved to be a less efficient way of indexing the peaks due to the weak intensity of the diffracted peaks and preferred orientation of the grains. Thus, to identify the phases present, the intensity was radially integrated in steps of 5° (for example, $0^\circ \leq \chi \leq 5^\circ, \dots, 175^\circ \leq \chi \leq 180^\circ$) and the peaks corresponding to the $\{hkl\}$ planes identified. The program Datasqueeze³⁷ was used to analyze the 2D data from the areal detector.

III. RESULTS

A. Grazing incidence x-ray diffraction

Figures 2 and 3 display GIXRD patterns, obtained using a point detector, from a 10 nm thick pure Ni film and a 10 nm thick $\text{Ni}_{0.95}\text{Pt}_{0.05}$ film, respectively, on an undoped Si(100) substrate in: the as-deposited condition; after RTA at 320°C for 5 s; after 420°C for 5 s; and after 420°C for 30 min. Nickel-silicide phases are heavily textured, and therefore using a point detector leads to the detection of only those Bragg reflections whose diffraction vectors are aligned with the detector.³⁸ To detect all the diffracting $\{hkl\}$ planes from a nickel-silicide film, we employed a 2D areal detector, which acquires data over a range of azimuthal angles ($0 \leq \varphi \leq 180$) with respect to the surface normal. Nonetheless, the point-detector patterns (Figures 2 and 3) are qualitatively representative of our main observations: (a) presence of Ni_2Si in the as-deposited condition; and (b) detection of NiSi and other nickel-rich phases on subjecting the sample to RTA heat treatments. In both the control and alloy samples, we detect the formation of the orthorhombic Ni_2Si phase³⁹ in the as-deposited films. Peaks corresponding to the nickel-monosilicide phase,⁴⁰ NiSi, are visible in both control

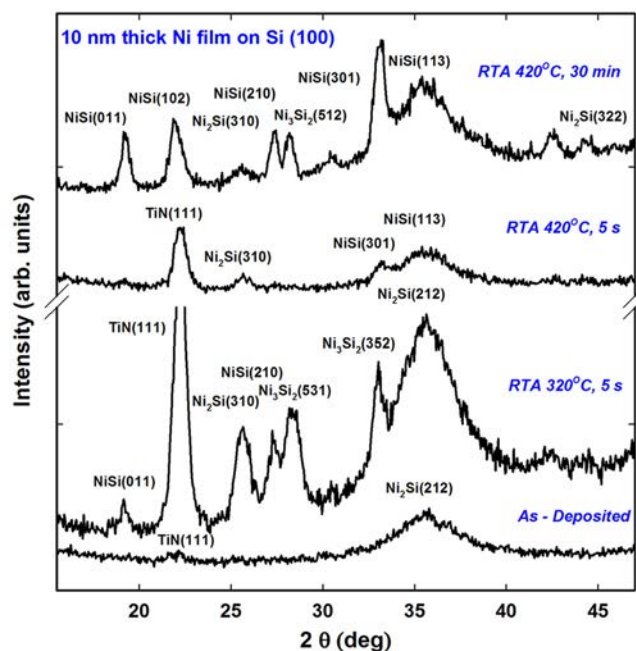


FIG. 2. Grazing-incidence XRD patterns recorded from a 10 nm thick Ni/Si(100) sample as-deposited and after RTA at 320°C (5 s) and 420°C (5 s and 30 min).

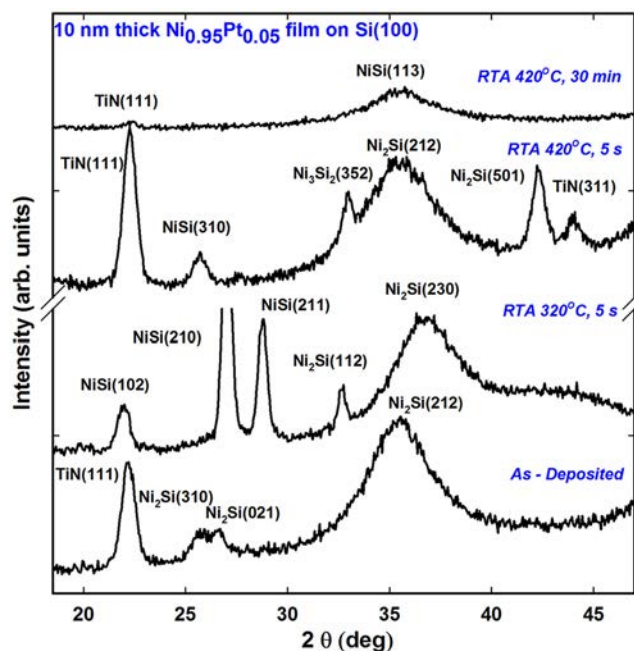


FIG. 3. Grazing-incidence XRD patterns recorded from a 10 nm thick $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ samples as-deposited and after RTA at 320°C (5 s) and 420°C (5 s and 30 min).

and alloy samples subjected to RTA at 320°C for 5 s, in addition to nickel-rich silicides,^{39,41,42} Ni_2Si and Ni_3Si_2 . Peaks corresponding to the TiN cap-layer, pure Ni and the Si substrate are also detected^{43–45} as anticipated. Quantification of the volume fraction of each phase is challenging due to strong texturing of the silicide phases; GIXRD measurements can only reliably be used to indicate the presence of a particular phase. The presence of NiSi at 320°C is remarkable as it is generally expected at temperatures $\geq 350^\circ\text{C}$. Samples subjected to RTA at 420°C for 5 s and 30 min also exhibit peaks corresponding to NiSi and the nickel-rich phases Ni_2Si and Ni_3Si_2 . This indicates that the reactive-diffusion mechanism resulting in NiSi from the nickel-rich phases is not fully complete at 30 min. The intensities of the Ni-rich peaks do, however, decrease with increasing annealing time from 5 s to 30 min.

B. Atom-probe tomographic measurements

Figure 4 shows a proximity histogram (or proxigram for short)⁴⁶ displaying the concentrations of Ni, Si, and Pt versus film depth for an alloy sample subjected to RTA at 320°C for 5 s. A proximity histogram is a 3D non-linear concentration profile, which is created by calculating the average composition of a series of isoconcentration surfaces as they are stepped through the thin-film. On the abscissa, zero indicates the nickel-silicide/Si(100) heterophase interface as denoted by the 75 at. % Si isoconcentration surface, which corresponds to the inflection point in the Si concentration profile. The inset figure shows the Ni-to-Si concentration ratio, in atomic percent, versus film-depth, which varies continuously from 2:1 (Ni_2Si) near the original surface to 1:1 (NiSi) near the NiSi/Si(100) interface. This indicates that the NiSi phase nucleates at the interface with Si(100). Also, the varying

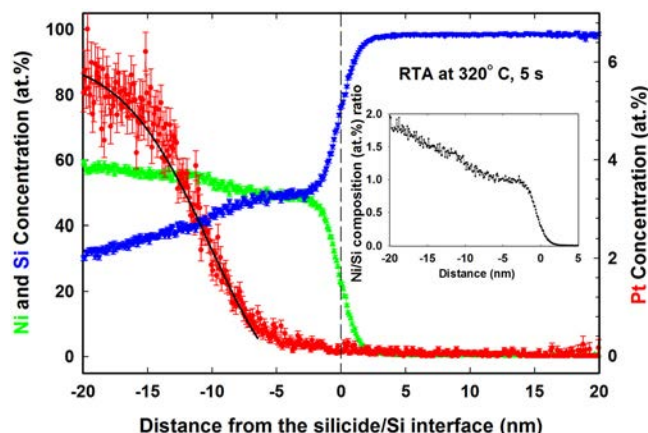


FIG. 4. Proximity histogram displaying the concentrations of Ni (green, $\blacktriangle$), Si (blue, $\blacktriangledown$), and Pt (red, $\bullet$) versus film depth for a 10 nm thick $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ film subjected to RTA at 320°C for 5 s. Concentrations were calculated for a series of 75 at. % Si isoconcentration surfaces moving into the thin film. On the abscissa zero represents the nickel-silicide/Si(100) heterophase interface. The black solid-line is a half-Gaussian fit to the Pt concentration profile. The inset diagram shows the Ni-to-Si concentration ratio (atomic percent) versus film depth.

Ni-to-Si concentration ratio is indicative of other nickel-rich phases, particularly Ni_3Si_2 (Ni/Si ~ 1.5). We report only concentrations from within the silicide portion of the film, where the Ga, Ti, and N (from the TiN cap) concentrations are less than a total of 2 at. %. Samples subjected to RTA at 420°C for 5 s exhibit a stoichiometric NiSi phase, ~ 5 nm thick, in contact with the Si(100) substrate, Figure 5. Near the $\text{Ni}_{1-x}\text{Pt}_x/\text{silicide}$ interface (far-left hand-side in Figure 5), the Ni-to-Si concentration ratio is still 2:1 (Ni ~ 60 at. %, Si ~ 30 at. %) and varies continuously until it achieves 1:1 in contact with the Si(100) substrate. It is clear that NiSi nucleates and grows *outwards* from the nickel-silicide/Si(100) heterophase interface. More of the nickel-rich silicides are consumed and converted to stoichiometric NiSi , as is evident by the growing NiSi phase in contact with the Si(100) substrate. The GIXRD

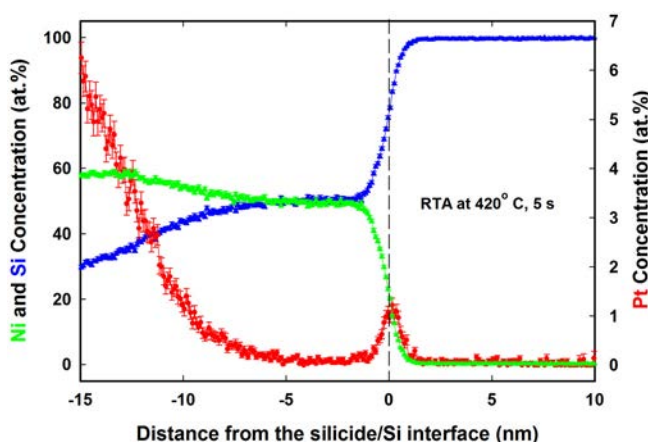


FIG. 5. Proximity histogram displaying the concentrations of Ni (green, $\blacktriangle$), Si (blue, $\blacktriangledown$), and Pt (red, $\bullet$) versus film depth for a 10 nm thick $(\text{Ni}_{0.95}\text{Pt}_{0.05})/\text{Si}(100)$ specimen subjected to RTA at 420°C for 5 s. Concentrations were calculated for a series of 75 at. % Si isoconcentration surfaces moving into the thin film. On the abscissa zero indicates the nickel-silicide/Si(100) interface.

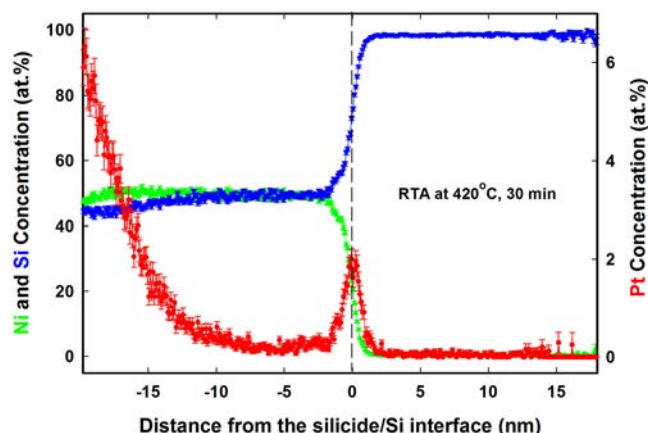


FIG. 6. Proximity histogram displaying the concentrations of Ni (green, $\blacktriangle$), Si (blue, $\blacktriangledown$), and Pt (red, $\bullet$) versus film depth for a 10 nm $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ thin film subjected to RTA at 420°C for 30 min. Concentrations were calculated for a series of 74 at. % Si isoconcentration surfaces moving into the thin film. Zero on the abscissa represents the nickel-silicide/Si(100) interface.

results are consistent with our LEAP tomography results and indicate the presence of NiSi , Ni_2Si , and Ni_3Si_2 phases in both the control and alloy samples. Figure 6 displays the proximity histogram of a $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ sample after 30 min at 420°C . The reaction-diffusion processes consuming the nickel-rich silicides, to form the NiSi phase, appear to be almost close to completion. The NiSi portion is ~ 13 nm thick. On the far-left hand-side, the Ni and Si concentrations still differ (the Ni concentration is greater than that of Si), which is indicative of the remnant nickel-rich silicide phases. There is no indication of the higher resistivity NiSi_2 phase near the $\text{NiSi}/\text{Si}(100)$ heterophase interface. As anticipated, diffraction peaks corresponding to NiSi are predominantly

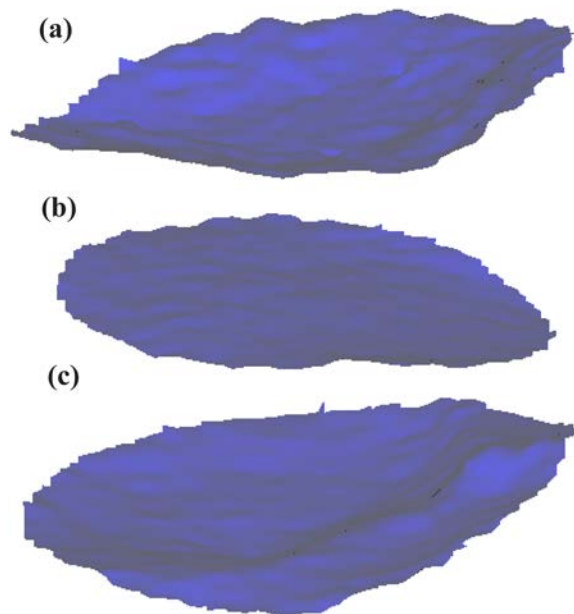


FIG. 7. Reconstruction of the silicide/Si(100) interface based on a 75 at. % Si isoconcentration surface for 10 nm thick $(\text{Ni}_{0.95}\text{Pt}_{0.05})$ film on Si(100) samples subjected to RTA at: (a) 320°C , 5 s; (b) 420°C , 5 s; and (c) 420°C , 30 min. The root-mean-square chemical roughnesses of the interfaces are 2.1, 0.6, and 1.2 nm, respectively.

observed employing GIXRD, along with some evidence for Ni_2Si and Ni_3Si_2 .

Figure 7 displays a 3D direct-space reconstruction of the silicide/Si(100) heterophase interface for a $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ sample after RTA at: (a) 320°C for 5 s; (b) 420°C for 5 s; and (c) 420°C for 30 min. The chemical RMS roughness of this heterophase interface based on a 75 at. % Si isoconcentration surface, which is a measure of the chemical inhomogeneity, is 2.1 nm after RTA at 320°C , which decreases to 0.6 nm at 420°C for 5 s, and correlates with the growth of a stoichiometric NiSi phase in contact with the Si(100) substrate, Figure 5. On further processing at 420°C for 30 min, the chemical RMS roughness (based on a 74 at. % Si isoconcentration surface) increases to 1.2 nm.

C. Platinum redistribution in $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ films

A half-Gaussian profile is observed for the Pt concentration versus film-depth (nm) when a $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ sample is subjected to RTA at 320°C for 5 s, Figure 4 (red, • curve); the solid black-line is a half-Gaussian fit to the Pt concentration profile. The peak Pt concentration is ~ 6 at. % near the Ni/silicide heterophase interface (far-left hand-side) and decreases to close to 0 at. % at the nickel-silicide/Si(100) heterophase interface. Figure 5 displays the Pt concentration profile (red, • curve) for the sample subjected to RTA at 420°C for 5 s. The peak in the Pt concentration profile (~ 6 at. %) occurs at the $\text{Ni}_{1-x}\text{Pt}_x/\text{silicide}$ interface (far-left hand-side). The Pt concentration decreases monotonically to ~ 0.12 at. % within the NiSi phase. Additionally, Pt is found to segregate at the NiSi/Si(100) heterophase interface. The non-equilibrium Pt Gibbsian interfacial excess is 1.2 ± 0.01 atoms nm^{-2} , see Sec. IV.

Figure 8(a) displays a $5 \times 30 \times 50$ nm³ slice of a 3D reconstruction from the nickel monosilicide portion of a $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ sample subjected to RTA at 420°C for 5 s. The atoms of elements Ni, Si, and Pt are indicated in green, blue, and red, respectively. The Pt atoms are decorating three GBs in the monosilicide ($\text{Ni}_{0.99}\text{Pt}_{0.01}$)Si phase. Additionally, Pt is detected at the top ($\text{Ni}_{1-x}\text{Pt}_x/\text{silicide}$) and bottom ($\text{Ni}_{0.99}\text{Pt}_{0.01}\text{Si}/\text{Si}$) heterophase interfaces. The ($\text{Ni}_{0.99}\text{Pt}_{0.01}$)Si grains have a columnar morphology. No gradient in the Pt concentration is found along the GBs, indicating very fast diffusion along them. Figure 8(a) also displays the $4 \times 4 \times 48$ nm³ cylindrical analysis-volume used to calculate the 1D concentration profiles across the three monosilicide grains, Figure 8(b). The Pt concentration profile has local maxima at the GBs of the nickel-monosilicide phase. A concomitant decrease in Ni concentration occurs at the GBs, Figure 8(b), resulting in local minima.

More Pt is found to segregate at the NiSi/Si(100) interface on further annealing at 420°C for 30 min, simulating the BEOL process, Figure 6. The Gibbsian interfacial excess of Pt at the NiSi/Si(100) heterophase interface is 2.1 ± 0.02 atoms nm^{-2} . Similar to the 5 s anneal at 420°C , the Pt concentration decreases from a maximum of 6 at. % near the $\text{Ni}_{1-x}\text{Pt}_x/\text{silicide}$ interface (far-left hand-side) to approximately 0.34 at. % within the NiSi portion of the film, before segregating at the NiSi/Si(100) interface, Figures 5 and 6.

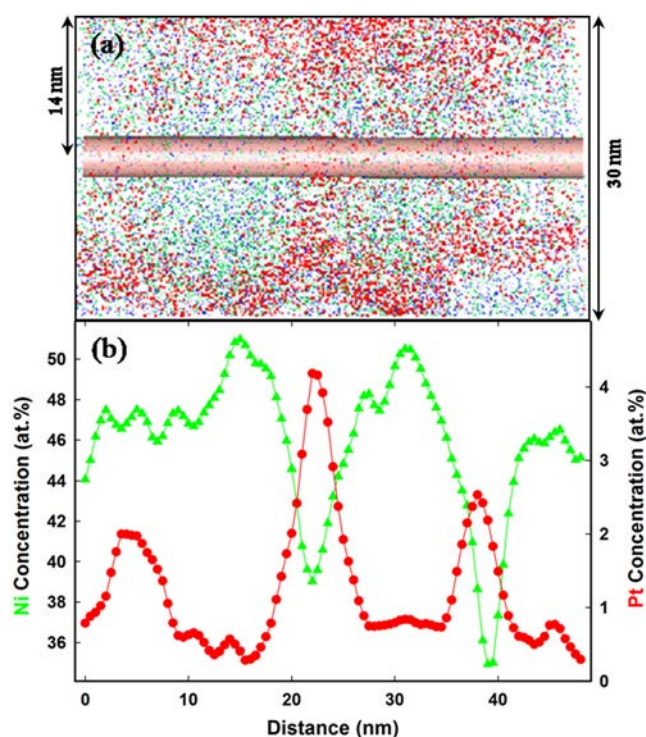


FIG. 8. (a) 2D vertical slice along the analysis direction of 3D LEAP tomographic data. An analysis volume of $30 \times 50 \times 5$ nm³ containing 2.88×10^5 atoms is displayed with Ni, Si, and Pt atoms represented in green, blue, and red, respectively. Only 10% of the Ni and Si atoms are displayed for clarity. (b) 1D composition profile of Pt and Ni within the $4 \times 4 \times 48$ nm³ cylindrical volume element placed horizontally across three GBs of the ($\text{Ni}_{0.99}\text{Pt}_{0.01}$)Si grains as displayed in (a). Reprinted with permission from P. Adusumilli, L. J. Lauhon, D. N. Seidman, C. E. Murray, O. Avayu, and Y. Rosenwaks, Appl. Phys. Lett. **94**, 113103 (2009). Copyright 2009 American Institute of Physics.

IV. DISCUSSION

A. Phase evolution of silicides

The reaction between Ni and Si is characterized by a complex phase-formation sequence, which depends on many processing parameters³ such as temperature, Ni film-thickness, dopant concentration, annealing conditions, and surface preparation. Understanding this phase evolution sequence is crucial for controlling the terminal phase, and thus the resistivity of a thin-film. Furthermore, the nucleation and growth of different silicide phases influence the diffusivity of Pt, and therefore the eventual distribution of Pt within the thin-film structure. Vice versa, the presence of Pt also affects silicide grain growth (Sec. IVB). Thus, it is important to understand the formation of the first phases forming as they influence the appearance of subsequent phases. We observe the presence of a crystalline Ni_2Si phase in the as-deposited thin-films along with unreacted $\text{Ni}_{1-x}\text{Pt}_x$, both with and without platinum, Figures 2 and 3. This is consistent with prior reports⁴⁷ on the formation of thin NiSi (~ 0.1 nm) and Ni_2Si (~ 1.4 nm) layers during vacuum evaporation of Ni onto Si(100) at room temperature. Any additional Ni atoms are reported to deposit as a pure Ni over layer. Intermixing between sputtered metal and silicon at or near ambient temperatures has been discussed

previously.^{48,49} For films deposited via sputtering the contribution of the kinetic energy of the sputtered ions needs to be considered.⁷ Some recent APT reports^{20,50,51} indicate the formation of Ni_2Si and NiSi in 80 nm thick as-deposited $\text{Ni}_{0.95}\text{Pt}_{0.05}$ films. Another crucial factor influencing the nucleation of the first silicide phase⁸ is substrate surface preparation. Most semiconductor systems go through a pre-heat cycle to remove moisture and physisorbed atoms from the substrate. In our case, the $\text{Si}(100)$ substrate was heated to $\sim 200^\circ\text{C}$ and then cooled prior to Ni film deposition at a base pressure of $\sim 1.6 \times 10^{-7}$ Torr; this is the lower bound of the range $200\text{--}350^\circ\text{C}$, in which Ni_2Si is typically anticipated in $\text{Ni}/\text{Si}(100)$ thin-films.⁵² If substrates are not cooled completely to room temperature prior to Ni deposition, then nucleation and growth of phases anticipated at temperatures slightly above room temperature is possible. The substrate preparation step appears to be the most important factor in our case and is responsible for the crystalline Ni_2Si phase in as-deposited films.

Rapid thermal processing of these samples at 320°C leads to nucleation of another nickel-rich silicide, Ni_3Si_2 , and the low-resistivity NiSi phase, according to our GIXRD measurements, Figures 2 and 3. This fact is consistent with our LEAP tomographic data, and is evident from the gradient in the Ni-to-Si atomic percent concentration ratio curve (inset of Figure 4), which varies continuously from 2:1 near the $\text{Ni}_{1-x}\text{Pt}_x/\text{silicide}$ heterophase interface to 1:1 at the $\text{NiSi}/\text{Si}(100)$ heterophase interface. Nickel-rich silicides Ni_2Si ($\text{Ni}/\text{Si} \sim 2$), Ni_3Si_2 ($\text{Ni}/\text{Si} \sim 1.5$), and the monosilicide NiSi ($\text{Ni}/\text{Si} \sim 1$) have concentration ratios that fall in this range. It is important to remember that the proximity histogram averages over the entire field-of-view, $50 \times 50 \text{ nm}^2$. Alternatively, the continuous gradient in the Ni and Si profiles might be explained by the presence of a $\theta\text{-Ni}_2\text{Si}$ phase, which is known to exist over a broad compositional range.⁵³ Although we do not observe any diffraction peaks corresponding to this metastable phase, a pole figure analysis is required to definitively rule out its existence due to peak overlap with other Ni-rich silicides.⁵⁴ The silicidation of a 10 nm thick $\text{Ni}_{0.95}\text{Pt}_{0.05}$ thin-film is expected to result in a thicker silicide film, whose thickness depends on the particular silicide phase(s) formed. When fully reacted to form NiSi , we expect a 22 nm thick $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ thin-film.³ At 420°C after 5 s, the NiSi phase grows at the expense of the nickel-rich silicides, Figure 5. Thus, it is clear that the NiSi phase nucleates at the nickel silicide/ $\text{Si}(100)$ interface and grows outward toward the original $\text{Ni}_{1-x}\text{Pt}_x/\text{silicide}$ interface, Figures 4 and 5. In a typical CMOS fabrication process, BEOL manufacturing represents a significant thermal budget for source-drain contacts and is simulated by subjecting a sample to RTA at 420°C for 30 min. After 30 min at 420°C , most of the nickel-rich silicide phases are consumed to form NiSi , which is now $\sim 13 \text{ nm}$ thick. The Ni-to-Si concentration ratio, in atomic percent, near the $\text{Ni}_{1-x}\text{Pt}_x/\text{silicide}$ interface (far-left hand-side in Figure 6) is greater than 1. This indicates the remnants of the nickel-rich silicides and is consistent with our GIXRD measurements. NiSi is metastable in contact with $\text{Si}(100)$, and there are no indications of a further reaction with Si leading to the higher resistivity NiSi_2 phase at the $\text{NiSi}/\text{Si}(100)$ interface.

The challenges of dopant variability and control of junction dimensions have made low-temperature processing imperative for present day manufacturing. Hence, our observations demonstrate that temperatures around $\sim 420^\circ\text{C}$ (lower-end of NiSi formation window) are appropriate for the fabrication of single-phase NiSi source-drain contacts using RTA treatments. Additionally, the thickness of the NiSi region of the silicide-film can be controlled by annealing time at 420°C . Although a number of factors influence the phase evolution sequence and kinetics, none is more important than the amount of Ni available during silicidation.^{3,7} Continuously decreasing device dimensions place limitations on the thickness of the starting Ni film and poses a challenge for achieving the requisite end phase, NiSi . A recent article suggests a very different phase formation sequence and texture for thinner $\text{Ni}_{1-x}\text{Pt}_x$ films ($<10 \text{ nm}$) on silicon-on-insulator (100) substrates.⁷ In the size regime where bulk resistivity dominates contact resistance, the formation of NiSi will be critical for achieving low-resistance source-drain contacts. Instead of starting with thinner $\text{Ni}_{1-x}\text{Pt}_x$ films, it should be possible to control the thickness of the NiSi phase in contact with $\text{Si}(100)$ and etch away the nickel-rich silicides^{55,56} to form uniform, single-phase low-resistance contacts for future technological generations.

As feature sizes decrease, the properties of the $\text{NiSi}/\text{Si}(100)$ interface will play an increasingly important role in determining the total resistance of the contact and subsequently the parasitic source-drain resistance of the circuit. Particularly, charge scattering at the interface will dominate contact resistance. Thus, it is critical to characterize this interface and follow its evolution during the silicidation process. Sheet-resistance measurements and light-scattering are typically used¹² to measure the roughness of the silicide/Si interface, but they only provide indirect evidence and are limited by resolution ($0.5 \mu\text{m}$ ⁵⁷). Furthermore, it is difficult to differentiate between (a) roughening of the interface or (b) the nucleation and growth of another silicide phase at the interface using these techniques. In contrast, 3D LEAP tomography offers a direct method for evaluating the chemical roughness and the evolution of the silicide/ $\text{Si}(100)$ heterophase interface by allowing us to reconstruct the buried interface in direct space. Additionally, we can concurrently study any phase transformations at the interface. Figure 7 displays direct space visualizations of the silicide/ $\text{Si}(100)$ heterophase interface after RTA treatments at: (a) 320°C for 5 s; (b) 420°C for 5 s; or (c) 420°C for 30 min. The chemical RMS roughness of the silicide/ $\text{Si}(100)$ interface, based on a 75 at. % Si isoconcentration surface, decreases from 2.1 nm after RTA at 320°C for 5 s to 0.6 nm after RTA at 420°C for 5 s. The decrease in the chemical RMS roughness of the silicide/ $\text{Si}(100)$ heterophase interface on the formation of NiSi is consistent with expectations of a smoother $\text{NiSi}/\text{Si}(100)$ interface, due to the diffusion-controlled formation of NiSi .³ On increasing the annealing time to 30 min the chemical RMS roughness of the $\text{Ni}(\text{Pt})\text{Si}/\text{Si}(100)$ interface, based on a 74 at. % Si isoconcentration surface, increases to 1.2 nm. APT data show evidence of variability of grain size at the interface and the increase in chemical RMS roughness may be a manifestation of larger grains, whose boundaries

become non-planar with this interface with a greater thermal budget.

B. Effect of platinum redistribution on NiSi microstructure and thermal stability

Platinum plays an important role in enhancing the morphological and thermal stability of NiSi at temperatures greater than 650 °C.¹¹ Understanding Pt redistribution in the silicide film during the silicidation process yields unique insights into the role it plays in phase stabilization. Diffusion of Pt is affected by the simultaneous reaction between Ni and Si to form different nickel-silicide phases, especially due to the differences in the Pt diffusivity in different crystal structures.⁵⁸ Figure 4 is the Pt concentration profile after RTA at 320 °C for 5 s; Pt follows a half-Gaussian profile, as exemplified by the half-Gaussian fit (black solid line) to the concentration curve. The lateral distribution of Pt is found to be very uniform at this temperature.

Rapid thermal processing at 420 °C for 5 s results in Pt segregation at the NiSi/Si(100) heterophase interface, Figure 5. Pt segregation at this interface is responsible for the enhanced morphological stability of NiSi against agglomeration at temperatures greater than 650 °C as reported earlier.¹⁵ Agglomeration of pure NiSi thin-films on Si(100) is driven by a decrease in the interfacial Gibbs free energy, which is achieved by a reduction in the contact area between the two phases. The enhanced resistance of (Ni_{1-x}Pt_x)Si films against agglomeration has been attributed to the reduction in the NiSi/Si(100) interfacial Gibbs free energy by short-circuit diffusion of Pt to this interface and its subsequent segregation,⁵ as implied by the Gibbs adsorption (segregation) isotherm^{59–61}

$$\Gamma_s = \left(\frac{\partial \sigma}{\partial \mu_s} \right)_{T,P}, \quad (1)$$

where σ is the NiSi/Si(100) interfacial free energy and μ_s is the chemical potential of the solute Pt. The Gibbsian interfacial excess of platinum, Γ_s , is used to quantify the segregation of a solute species at a heterophase interface, Eq. (2). The proximity histogram yields a discrete count of the atoms in the vicinity of a heterophase interface, thus, permitting a direct determination of Γ_s without the need for a Gibbs dividing surface⁶²

$$\Gamma_s = \sum_n \left[\frac{(c_n - c_o)}{A_l} \right] / (1 - c_o) \quad (2)$$

and $A_l = N_l / \rho \Delta l$, where A_l is the effective area of a slice at proximity l , N_l is the number of atoms in the slice, Δl is the shell thickness, and ρ is the ideal atomic density of the monosilicide phase. c_n is the concentration of the n th atom (unity if it is the segregating species of interest and zero otherwise) and c_o is the bulk concentration of that species. The non-equilibrium Pt Gibbsian interfacial excess at the NiSi/Si(100) interface, after RTA at 420 °C for 5 s, is 1.2 ± 0.01 atoms nm⁻². On increasing the annealing time at 420 °C to 30 min (Figure 6), the Gibbsian interfacial excess of Pt at the

NiSi/Si(100) interface increases to 2.1 ± 0.02 atoms nm⁻². The Gibbsian interfacial excess can be used to calculate the reduction in the interfacial Gibbs free energy as a result of segregation. Employing Henry's law and assuming a constant value for the Henry's law constant for a given Pt solute concentration equation (1) becomes

$$d\mu_{Pt} = k_B T dc_{Pt} / c_{Pt}, \quad (3)$$

where k_B is Boltzmann's constant, T is the absolute temperature (K) used for annealing, and c_{Pt} is the concentration of Pt. Combining Eqs. (1) and (3) yields

$$\frac{d\sigma}{dc_{Pt}} = \left(\frac{k_B T}{c_{Pt}} \right) \Gamma_{Pt}, \quad (4)$$

which yields

$$\left(\frac{1}{k_B T \Gamma_{Pt}} \right) \int_{\sigma_{initial}}^{\sigma_{final}} d\sigma = \int_{c_{initial}}^{c_{final}} \frac{dc}{c}. \quad (5)$$

And then integrating equation (5), one obtains the magnitude of the decrease in the interfacial free energy, $\Delta\sigma$, due to segregation

$$\Delta\sigma = (k_B T \Gamma_{Pt}) \ln \left(\frac{c_{final}}{c_{initial}} \right), \quad (6)$$

where Γ_{Pt} is the Gibbsian interfacial excess of Pt at the NiSi/Si(100) interface; $c_{initial}$ and c_{final} are the Pt concentration in the monosilicide phase, and the peak concentration at the NiSi/Si(100) heterophase interface, respectively. In our case, the average Pt concentration in the NiSi phase, $c_{initial}$ is ~ 1.40 at. % and the Pt peak concentration at the NiSi/Si(100) interface, c_{final} , is ~ 1.99 at. %. The coefficient of the reduction in interfacial free energy $d\sigma/dc_{Pt}$ (Eq. (4)) is 1005 mJ m⁻² (at. fr.)⁻¹. Therefore, segregation of Pt at the NiSi/Si(100) interface reduces its interfacial Gibbs free energy by 7.1 mJ m⁻². (We have reported¹³ similar segregation behavior at the (Ni_{0.98}Pd_{0.02})Si/Si(100) heterophase interface when Pd is used as an alloying addition.) Furthermore, the segregated Pt is found to be laterally uniform at the NiSi/Si(100) interface. Due to a higher electron work function than NiSi (~ 4.75),^{63,64} Pt (~ 5.12 – 5.64)⁶⁵ results in a smaller potential barrier in contact with p-type Si and has a beneficial effect on the contact resistance in p-type metal-oxide-semiconductor transistors.⁶⁶

C. Short-circuit diffusion of platinum along grain boundaries

Figure 8(a) displays a $50 \times 5 \times 30$ nm³ sub-volume of an APT reconstruction exhibiting the nickel monosilicide portion of the film in a sample subjected to RTA at 420 °C for 5 s. Figure 8(b) displays 1D concentration profiles of Pt and Ni atoms in a cylindrical analysis volume, placed ~ 14 nm below the Ni_{1-x}Pt_x/silicide interface, which is perpendicular to three (Ni_{0.99}Pt_{0.01})Si grains. Pt atoms (red dots) are clearly seen decorating the NiSi grain boundaries.¹⁴ Pt is found to accumulate not only at the GBs, but at the

$(\text{Ni}_{1-x}\text{Pt}_x)/(\text{Ni}_{0.99}\text{Pt}_{0.01})\text{Si}$ (top) and $(\text{Ni}_{0.99}\text{Pt}_{0.01})\text{Si}/\text{Si}(100)$ (bottom) heterophase interfaces as well, Figure 8(a). The cylindrical volume is placed at a depth greater than the bulk RMS diffusion distance of Pt (~ 3 nm) from the $(\text{Ni}_{1-x}\text{Pt}_x)/(\text{Ni}_{0.99}\text{Pt}_{0.01})\text{Si}$ heterophase interface. The significantly higher concentration of Pt in the GBs compared to the bulk of the grains implies that short-circuit diffusion of Pt is occurring via the GBs in NiSi, Figure 8(b). A concomitant decrease in the Ni concentration profile implies that Pt is substituting for Ni in the GBs.

There are two possible models to describe our experimental models: (1) Harrison's type-B model; or (2) the Hwang-Balluffi (H-B) model. We first consider Harrison's type-B model and then consider the H-B model, and finally argue that Harrison's type-B model is the better of the two models.

Our Pt concentration profiles are describable by Harrison's type-B regime^{58,60,67–70} model for short-circuit diffusion along high-diffusivity paths, that is, GBs. Also known as the "leaky-pipe" model, the type-B regime is characterized by: (a) some Pt volume diffusion from the unreacted $\text{Ni}_{1-x}\text{Pt}_x$ film into the $(\text{Ni}_{0.99}\text{Pt}_{0.01})$ grains; (b) some volume diffusion laterally from the GBs into the $(\text{Ni}_{0.99}\text{Pt}_{0.01})\text{Si}$ grains; and (c) predominately short-circuit diffusion along the monosilicide grain boundaries, as illustrated in Figure 9.

The Pt concentration profile within the $\text{Ni}_{0.99}\text{Pt}_{0.01}\text{Si}$ grains is a function of its grain size. For smaller grains, with diameters less than ~ 15 nm, the average Pt concentration is higher due to overlap of the Pt concentration profiles from adjacent GBs; Figure 8(a) (center grain). Thus, variability in grain diameter leads to variations in the average Pt concentration within the $\text{Ni}_{0.99}\text{Pt}_{0.01}\text{Si}$ grains. There is not, however, a measurable Pt concentration gradient along the GBs. The high spatial-resolution available in 3D enables direct and accurate measurement of lattice (D_L) and GB (D_{GB}) diffusivities at 420°C . A D_L of $4.2 \pm 1.2 \times 10^{-19} \text{ m}^2 \text{ s}^{-1}$ is obtained by plotting the $\log_{10}(c_{Pt})$ within the grains versus distance-squared, averaged over 10 grains.¹⁴ GB diffusivity, D_{GB} , is calculated using the Whipple-Suzuoka equation^{71,72}

$$\left(\frac{d\ln c_{Pt}}{dy^{6/5}}\right)^{5/3} = 0.66 \left(\frac{D_L}{t}\right)^{1/2} / \delta D_{GB}, \quad (7)$$

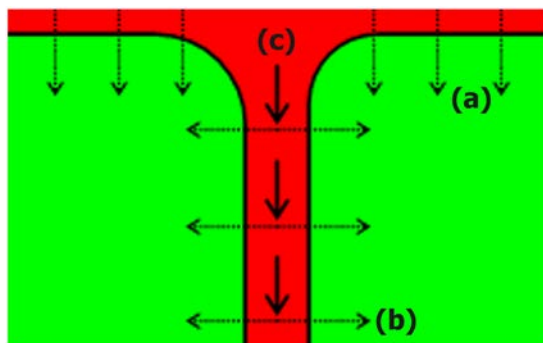


FIG. 9. Schematic diagram depicting Harrison type-B kinetics: (a) volume diffusion into the bulk from the unreacted $\text{Ni}_{1-x}\text{Pt}_x$ metal on top surface; (b) volume diffusion laterally from the grain boundaries; and (c) short-circuit diffusion along the grain boundaries.¹⁵

where c_{Pt} is the concentration of Pt within the $(\text{Ni}_{0.99}\text{Pt}_{0.01})\text{Si}$ grain; t is the time of the anneal (5 s); y is the distance perpendicular to the $(\text{Ni}_{1-x}\text{Pt}_x)/(\text{Ni}_{0.99}\text{Pt}_{0.01})\text{Si}$ interface; and δ is the width of the GB, which is taken to be 0.5 nm.⁷³ The Whipple-Suzuoka equation assumes a single-phase, semi-infinite system, and we make the same assumption. Utilizing the Whipple-Suzuoka equation, a D_{GB} of $1.5 \pm 0.2 \times 10^{-16} \text{ m}^2 \text{ s}^{-1}$ is obtained, which is ~ 350 times greater than D_L . This establishes quantitatively GB diffusion as the predominant mass transfer mechanism for Pt within the silicide thin film. These diffusivity measurements are, however, affected strongly by the simultaneous reaction between Ni and Si to form nickel monosilicide, which is not included in our analysis and is a more general problem that remains to be solved.

The Hwang-Balluffi (H-B) analysis^{74,75} of short-circuit diffusion along grain boundaries is for thin-films residing on a substrate, followed by fast diffusion along the thin-film/substrate interface. The H-B analysis assumes Harrison-type C short-circuit diffusion, which implies there is *no leakage of solute atoms into the grains*, whereas our observations indicate predominately Harrison-type B short-circuit diffusion because there is some leakage of solute atoms into the grains. Our APT experiments exhibit the following results, which are consistent with Harrison-type B short-circuit diffusion: (1) there is leakage of Pt from the grain boundaries into the grains as demonstrated by the Pt concentration profiles within them; (2) the level of the Pt concentrations within the grains is a function of grain diameter; (3) with respect to the latter point, the smaller the grain diameter the higher is the Pt concentration within the grains, which is physically reasonable since the RMS diffusion distance is shorter in small grains and hence in a given time there will be more Pt atoms residing within them. These three points demonstrate that our experiments exhibit mainly Harrison-type B short-circuit diffusion of Pt and not H-B. The fact that we could not measure a significant Pt concentration gradient along the GBs does not rule out Harrison-type B short-circuit diffusion. This conclusion is also important for interpreting the Pt accumulation at the NiSi/Si interface. Fast diffusion via short-circuit paths can cause solute accumulation at the bottom interface of a polycrystalline thin-film, even in the absence of segregation. It can be shown, however, that in the Harrison-type B kinetic regime, this accumulation profile is broad because of solute diffusion from the interface to the bulk of the grains.⁷⁶ In our case, the sharp concentration peak at the interface demonstrates that this Pt accumulation can only be caused by interfacial segregation and not only by diffusion. Finally, the observation that the chemical RMS roughness of the NiSi/Si interfaces decreases with increasing annealing time also implies that our observations of an excess Pt concentration at this interface must be due to segregation of Pt, which is a result, according to the Gibbs adsorption (segregation) isotherm, of a decrease in the interfacial Gibbs free energy. If the excess Pt we measured at the NiSi/Si interface was solely due to diffusion of Pt (H-B C-type diffusion along grain boundaries, followed by fast diffusion along the NiSi/Si interface) there would not be a decrease in the chemical RMS roughness of this interface

because segregation is thermodynamically driven by a decrease in interfacial free energy, which results in “wetting” of the NiSi/Si interface.

The observation of short-circuit diffusion of Pt via GBs provides unique insights into the kinetics of diffusion during silicidation and has important implications for NiSi formation and retention. The presence of high-diffusivity paths enables Pt to diffuse rapidly to the NiSi/Si(100) interface during the early stages of NiSi nucleation and growth at 420 °C. This may explain why alloying with Pt modifies the NiSi texture. The presence of Pt at the NiSi GBs affects the subsequent growth of NiSi grains due to a solute-drag effect⁷⁷ and thus leads to smaller grain diameters.⁷⁸ Additionally, (Ni_{0.99}Pt_{0.01})Si grains exhibit a columnar morphology. How these changes in the microstructure affect electrical properties is not yet completely understood. As the grain diameter decreases electron scattering from GBs becomes more important because the GB area to grain volume ratio increases.^{79,80} For the sample subjected to 420 °C for 30 min to simulate the BEOL process, we did not find any GBs within the APT reconstruction. On increasing the annealing time to 30 min, additional Pt will, however, diffuse into the monosilicide grains via lattice diffusion of Pt from both the Ni_{1-x}Pt_x/silicide heterophase interface and the GBs.

V. CONCLUSIONS

Local-electrode atom-probe tomography and grazing-incidence x-ray diffraction measurements were used to study in detail the formation and temporal evolution of nickel-silicide phases, and the redistribution of Pt during the temporal evolution of silicidation of 10 nm thick (Ni_{0.95}Pt_{0.05}) thin-films. These films were studied as-deposited and after RTA treatments at 320 and 420 °C. Thus, we are studying the kinetics of the formation of NiSi, which reveals new insights into silicidation. The following conclusions are reached:

- (1) The nickel-rich Ni₂Si phase found in the as-deposited films can be attributed to surface preparation prior to sputter deposition, Figures 2 and 3.
- (2) Ni₂Si is consumed to form a nickel-rich silicide Ni₃Si₂, and the NiSi phase at the silicide/Si(100) interface, after RTA at 320 °C for 5 s, Figures 2, 3, and 4. Thus, our experiments demonstrate the kinetics of formation of NiSi.
- (3) NiSi continues to grow outward toward the Ni_{1-x}Pt_x/silicide interface by the consumption of the nickel-rich phases at 420 °C, Figures 5 and 6. The precise control of the thickness of the monosilicide phase is possible by varying the annealing time at this temperature.
- (4) The nickel-silicide/silicon heterophase interface is reconstructed in 3D and its chemical root-mean-square roughness is found to decrease to 0.6 nm with the formation of NiSi, Figure 7.
- (5) Short-circuit diffusion of Pt via NiSi grain-boundaries is directly observed. This provides mechanistic insights into the diffusion of Pt to and subsequent segregation at the NiSi/Si(100) heterophase interface, Figure 8. The lattice (D_L) and grain boundary (D_{GB}) diffusivities are calculated to be $4.2 \pm 1.2 \times 10^{-19}$ and $1.5 \pm 0.2 \times 10^{-16}$ m² s⁻¹, respectively, after annealing at 420 °C for 5 s: based on the Whipple-Suzuoka equation assuming Harrison type-B description of short-circuit diffusion. This implies that D_{GB} is approximately 357 times greater than D_L .
- (6) It is argued that the Hwang-Balluffi model, which assumes Harrison type-C short-circuit diffusion, does not adequately describe our APT observations.
- (7) The enhanced resistance of (Ni_{1-x}Pt_x)/Si films against agglomeration is attributed to the reduction of the NiSi/Si(100) interfacial Gibbs free energy as a result of segregation of Pt at this interface. The Gibbsian interfacial excess of Pt at the NiSi/Si(100) interface after RTA at 420 °C for 5 s is 1.2 ± 0.01 atoms nm⁻² and after 30 min it is 2.1 ± 0.02 atoms nm⁻², Figures 5 and 6.
- (8) Pt short-circuit diffusion via grain boundaries in the NiSi grains significantly affects the grain diameter, morphology, and possibly the preferred orientation of NiSi grains. The Ni(Pt)Si grains have a columnar morphology and smaller average grain diameter when compared to an unalloyed NiSi thin film, Figure 8.

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